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Session Tracking Techniques

CSA4305– Internet Programming

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INTRODUCTION



- **HTTP is a stateless protocol**, meaning each request is treated independently by the web server.
- **Session tracking** is used to maintain a user's information across multiple requests.
- It helps web applications remember details such as **login status, user preferences, and shopping cart items**.
- Java web applications provide several session tracking mechanisms.
- The three important techniques are:
 - **HttpSession**
 - **Cookies**
 - **URL Rewriting**



What is Session Tracking?



- ❑ HTTP is a **stateless protocol**, meaning the server does not automatically remember previous requests.
- ❑ Session tracking is used to **maintain information about a user across multiple requests**.
- ❑ It helps web applications identify users and preserve their activities.
- ❑ Examples:
 - ❑ User login status
 - ❑ Shopping cart items
 - ❑ User preferences
 - ❑ Online banking sessions



Why Session Tracking is Needed?



- ❑ Maintains user identity after login.
- ❑ Stores temporary user information.
- ❑ Provides personalized content.
- ❑ Maintains shopping cart information.
- ❑ Prevents users from repeatedly entering information.
- ❑ Supports secure and continuous user interaction.
- ❑ **Example:**
 - Login → Browse Products → Add to Cart → Checkout**
- ❑ The application needs to know that all these requests belong to the **same user**.



HttpSession



□ HttpSession Mechanism

□ HttpSession is a server-side session management mechanism provided by Java Servlets.

□ How it works:

- User sends a request.
- Server creates or retrieves an HttpSession.
- A unique **session ID** is assigned.
- Session data is stored on the server.
- The session ID is sent back to the client.
- The client sends the session ID with subsequent requests.



Advantage HttpSession



- ❑ Session data is stored on the **server**.
- ❑ Can store multiple types of user information.
- ❑ Easy to use with Java Servlets.
- ❑ Suitable for login and authentication.
- ❑ Client does not directly store the actual session data.



Limitations



- ❑ Consumes server memory.
- ❑ Large numbers of users can increase server resource usage.
- ❑ Requires session management and expiration.
- ❑ Session ID theft can lead to session hijacking.

Best Use

- ❑ **Login systems, shopping carts, banking and personalized applications.**



Cookies



❑ **Cookie-Based Session Tracking**

- ❑ A cookie is a small piece of data stored in the user's browser.
- ❑ **How it works:**
 - ❑ Server sends a cookie to the browser.
 - ❑ Browser stores the cookie.
 - ❑ Browser automatically sends the cookie with future requests.
 - ❑ Server uses the cookie to identify the user/session.



URL REWRITING



URL Rewriting Mechanism

- URL rewriting maintains the session by adding the **session ID to the URL**.

Example:

- Normal URL:
- `www.example.com/products`
- Rewritten URL:
- `www.example.com/products;jsessionid=ABC123`

How it works:

- User requests a page.
- Server creates a session.
- Session ID is added to links.



Session Persistence Comparison



Mechanism	Where Data/ID is Maintained	Persistence
HttpSession	Server	Until timeout/logout
Cookies	Client browser	Session or persistent cookie
URL Rewriting	Session ID in URL	Usually session-based



Security Comparison



Mechanism	Main Security Concern	Protection
HttpSession	Session hijacking	HTTPS, session timeout, session ID regeneration
Cookies	Cookie theft/XSS/CSRF	Secure, HttpOnly, SameSite, HTTPS
URL Rewriting	Session ID exposure	HTTPS, avoid exposing sensitive IDs, proper session management



Use-Case Comparison



Use Case	Recommended Mechanism
User Login	HttpSession + secure cookie
Shopping Cart	HttpSession
Remember User Preferences	Cookies
Cookie Disabled	URL Rewriting
Banking Application	HttpSession + HTTPS
Simple Website Preferences	Cookies
Java Servlet Application	HttpSession



Conclusion



- Session tracking is essential because HTTP is stateless.
- **HttpSession** is suitable for maintaining server-side user sessions.
- **Cookies** are useful for storing small client-side information and session identifiers.
- **URL rewriting** provides an alternative when cookies are unavailable.
- Each mechanism has different advantages and security limitations.

THANK YOU